

TLH- 12120 : Enhance Org Onboarding to Maintain Org Config Version and Publish OrgConfigPublished Event

1. Purpose

Existing Org Onboarding should be enhanced to maintain a versioned **Org Config** for each org. This config captures what the org is enabled to use at account/setup level.

This is not a new onboarding service. It is an enhancement to the current org onboarding flow.

Metadata Registry remains the global source for reusable metadata values and does not store org-specific enabled values.

2. User Story

As a **Platform Admin / Org Onboarding Admin**,

I want Org Onboarding to save and version the org's enabled modules, programs, vitals, devices, languages, and linked org setup,
so that downstream services can bootstrap or update their own org-specific configuration using a consistent org setup source.

3. Core Principle

Org Onboarding owns org setup/config.

Each service owns its own org-scoped data.

OrgConfigPublished tells services that org config changed.

Services call Org Config API to fetch full details.

OCP tracks readiness later; it is not the setup data source.

Consuming services should store their own configuration and metadata-code references, while OCP remains a readiness/status projection.

4. Scope

In Scope

Area	Included
Existing org onboarding enhancement	Add org config versioning to current onboarding flow
Org setup data	Modules/features, country, language, categories/conditions, vitals/devices, linked orgs, agreements/forms
Metadata validation	Validate selected codes against Metadata Registry
Config versioning	Maintain current org config version and previous versions for audit
Lightweight event	Publish OrgConfigPublished when config is confirmed/changed
API access	Downstream services fetch full config from Org Config API
Audit	Track changed sections, changed by, changed at, version, and reason

5. Org Config Version — Data to Store

Maintain a current org config version as part of Org Onboarding / Org Config.

Field	Required	Notes
OrgID	Yes	Org being configured
OrgConfigVersion	Yes	Incrementing version
IsCurrentVersion	Yes	True for active/latest version
OrgName	Yes	From org profile
CountryCode	Yes	Metadata-backed
Timezone	Yes	Selected/derived
DefaultLanguageCode	Yes	Metadata-backed
SupportedLanguageCodes	Optional	Multi-select
EnabledModuleCodes	Yes	Appointment, RPM, Package, CarePlan, Payment, etc.
EnabledFeatureCodes	Optional	VideoVisit, Chat, PaymentLink, etc.
EnabledCategoryCodes	Optional	Chronic, Wellness, PostOp, General
EnabledConditionCodes	Optional	Hypertension, Diabetes, Knee Surgery, etc.
EnabledMetricCodes	Optional	BP, glucose, weight, SpO2, etc.
EnabledDeviceCodes	Optional	BP monitor, glucometer, weighing scale, etc.
LinkedOrgReferences	Optional	Labs, pharmacy, partner orgs
RequiredAgreementIDs	Optional	Documents/forms/agreements
ChangedSections	Yes	Modules, Programs, Vitals, Devices, Partners, etc.
PublishedBy / PublishedAt	Yes	Audit
ChangeReason	Optional	Why config changed

No separate business status is needed here. Existing Org Onboarding status remains the business workflow status.

5.1. Org Capability References

- Org Config shall create and maintain OrgCapabilityID values for each enabled org capability that requires downstream readiness tracking.
- OrgCapabilityID is created and owned by Org Config / Org Onboarding when the capability is first enabled for the org.
- OCP shall not create OrgCapabilityID. OCP shall only consume OrgCapabilityID from Org Config and track readiness/status against it.
- Each published Org Config version shall include the OrgCapabilityID values and their status for that version.
- OrgCapabilityID history shall be managed through Org Config versions.
- When a capability remains the same across Org Config versions, the same OrgCapabilityID shall be reused.
- When a capability is disabled, the same OrgCapabilityID shall remain in the version payload with CapabilityStatus = Disabled.
- When a new capability is enabled, Org Config shall create a new OrgCapabilityID.
- Org Config APIs shall return OrgCapabilityID values as part of the current and versioned Org Config payload.

Examples:

- Chronic + Hypertension

- BP Monitoring
- Video Consult
- Payment Link

Field	Required	Notes
OrgCapabilityID	Yes	Stable system-generated ID created by Org Config
CapabilityType	Yes	Program / Monitoring / Feature / Payment / Device / Partner
CapabilityDisplayKey	Yes	Human-readable/debug key, e.g. CHRONIC:HYPERTENSION, MONITORING:BP, FEATURE:VIDEO_VISIT
CapabilityStatus	Yes	Enabled / Disabled

OrgCapabilityID	CapabilityType	CapabilityDisplayKey	CapabilityStatus
CAP-001	Program	CHRONIC:HYPERTENSION	Enabled
CAP-002	Monitoring	MONITORING:BP	Enabled
CAP-003	Feature	FEATURE:VIDEO_VISIT	Enabled

6. Metadata Validation Rules

Before confirming/publishing org config changes:

All selected metadata-backed codes must be active and valid in Metadata Registry.

Org-specific enablement is stored in Org Config, not Metadata Registry.

Metadata Registry is used for validation and reference lookup only.

Examples:

Org Config Field	Validate Against
CountryCode	Country metadata
LanguageCode	Language metadata
EnabledModuleCodes	Module / Feature metadata
EnabledCategoryCodes	Category metadata
EnabledConditionCodes	Condition metadata
EnabledMetricCodes	Metric / Vital metadata
EnabledDeviceCodes	Device metadata

7. Bootstrap / Update Flow

- Platform Admin updates existing Org Onboarding setup.
- Org Config validates selected metadata-backed codes.
- Org Config saves a new OrgConfigVersion.
- New version becomes current.
- Previous current version is retained for audit.
- Org Config publishes lightweight OrgConfigPublished event.
- Downstream services receive event.
- Each service calls Org Config API to fetch full current config if needed.
- Each service creates/updates its own org-scoped data.
- Each service reports readiness to OCP using readiness signal contract.

- Org Config creates or updates Org Capability records for enabled capabilities that require downstream readiness tracking.
- Org Config includes OrgCapabilityID values in the current org config payload available through Org Config APIs.

8. Event

Use **one event**.

OrgConfigPublished

Purpose: notify downstream services that org config is current/changed and may require bootstrap or re-evaluation.

Do **not** include full org setup payload in the event.

Field	Required	Notes
EventID	Yes	Unique event
EventType	Yes	OrgConfigPublished
OrgID	Yes	Org
OrgConfigVersion	Yes	Current version
ChangeType	Yes	Initial / Update / Rebootstrap / Disable
ChangedSections	Yes	Modules, Programs, Vitals, Devices, Languages, Partners, Agreements
TargetServices	Optional	Template, ServiceCatalog, Package, Appointment, RPM/Device, Payment, Partner, Forms
PublishedAt	Yes	Timestamp
PublishedBy	Yes	Actor/system
CorrelationID	Optional	Trace/debug

Services should call Org Config API for the full payload.

9. Downstream Service Behavior

Each service should:

Step	Behavior
1	Consume OrgConfigPublished
2	Check whether ChangedSections or TargetServices apply
3	Call Org Config API if full setup details are needed
4	Create/update its own org-scoped configuration
5	Report readiness to OCP

Examples: (OR as Required)

Service	Expected Behavior
Template Service	Bootstrap/update org masters/templates/care plans for enabled program contexts
Service Catalog	Enable service setup options or default services for enabled org context

Package Service	Enable package setup shell/options
Appointment Service	Enable appointment/visit configuration
RPM/Device Service	Store enabled vitals/devices/source rules
Payment/Billing	Initialize checkout/payment config
Partner/Linked Org	Store linked lab/pharmacy/partner context

10. APIs Required

Get Current Org Config

GET /orgs/{orgId}/config/current

Used by downstream services after receiving OrgConfigPublished.

Response shall include OrgCapability records and OrgCapabilityID values for capabilities enabled in the current Org Config version.

Get Org Config by Version

GET /orgs/{orgId}/config/versions/{version}

Used for audit/debug/reprocessing.

Response shall include OrgCapability records applicable to that Org Config version for audit and repr

Update Org Config

PUT /orgs/{orgId}/config

Used by existing onboarding/admin flow to update org setup.

Publish Org Config

POST /orgs/{orgId}/config/publish

Validates config, creates new current version, and emits OrgConfigPublished.

11. Status / Version Rules

Use existing **Org Onboarding Status** for business workflow.

Org Config only needs versioning:

CurrentVersion

PreviousVersion

PublishedAt

PublishedBy

ChangeReason

Rules:

Only the current Org Config version is used for new downstream setup.

Previous versions are retained for audit.

No separate snapshot status UI is required.

Admin UI may show current config version and last published date for debug/audit only.

12. Acceptance Criteria

Scenario 1 — Update org config from onboarding

Given an existing org onboarding record,

When admin updates modules, programs, vitals, devices, languages, or linked orgs,

Then Org Onboarding shall save the updated setup as a new Org Config version.

Scenario 2 — Validate metadata-backed fields

Given org config contains metadata-backed values,

When admin publishes the config,

Then system shall validate selected codes against active Metadata Registry values.

Scenario 3 — Publish lightweight event

Given org config is successfully published,

When the new version becomes current,

Then system shall publish OrgConfigPublished with OrgID, OrgConfigVersion, ChangeType, ChangedSections, TargetServices, and CorrelationID.

Scenario 4 — Event does not carry full config

Given OrgConfigPublished is emitted,

When downstream services receive the event,

Then the event shall not include the full org setup payload,

And services shall call Org Config API when full config is needed.

Scenario 5 — Downstream service owns its setup

Given Template, Package, Service Catalog, Appointment, RPM, Payment, or Partner service receives the event,

When service needs to update org-scoped setup,

Then the service shall update its own data,

And shall not rely on OCP for detailed org config.

Scenario 6 — Previous version retained

Given a new Org Config version is published,

When it becomes current,

Then previous version shall remain available for audit,

And only the latest current version shall be used for new bootstrap/update processing.

Scenario 7 — No separate snapshot status UI

Given Org Config versioning exists,

When admin views onboarding status,

Then UI shall use existing Org Onboarding status,

And may show Org Config version only as audit/debug information.

Scenario 8 — OrgCapabilityID creation

Given admin enables a capability that requires readiness tracking,
When Org Config is published,
Then Org Config shall create or reuse an OrgCapabilityID for that capability,
And the OrgCapabilityID shall be included in the current org config API response,
And OCP shall use that OrgCapabilityID for readiness tracking.

Scenario 9 — OCP does not create OrgCapabilityID

Given OrgConfigPublished is emitted,

When OCP processes the change,
Then OCP shall read OrgCapabilityID from Org Config,
And OCP shall not generate its own OrgCapabilityID.